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 Website: www.ktunotes.in

- c) Sequence Control – Two or more stages of voltage controllers are used. They are triggered in proper sequence one after the other so as to obtain a variable harmonic content

Applications : Industrial heating, induction heating, speed control of induction motor, power control, lighting control etc.

PHASE ANGLE CONTROL

Switches connect the load to the ac source for a portion of each cycle of input voltage.

Thyristors are line commutated; no need of complex commutation circuit.

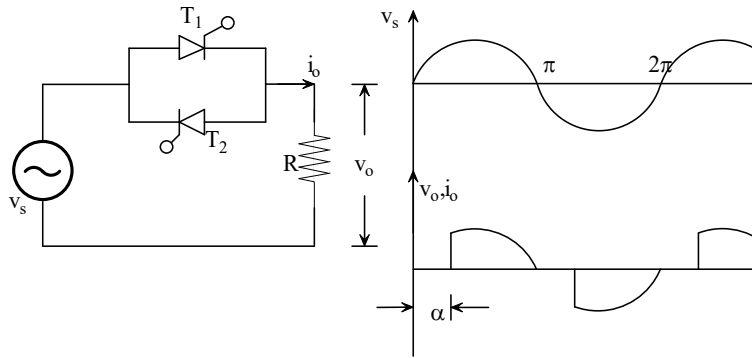
Disadvantage: Presence of harmonics in the supply current and load voltage

1-Phase Full Wave AC voltage controller

Here, two thyristors are used in anti-parallel.

R LOAD

T_1 and T_2 are forward biased during positive and negative half cycles respectively. In the positive half cycle, T_1 is triggered at a firing angle α . T_1 starts conducting. $v_o = v_s$ from α to π and i_o fall to zero (anode current below holding current) and T_1 is turned off. In the negative half cycle, T_2 is triggered at $\pi + \alpha$. T_2 conducts from $\pi + \alpha$ to 2π . At 2π , T_2 is turned off.



Here, positive and negative areas of v_o are same; hence no dc component.

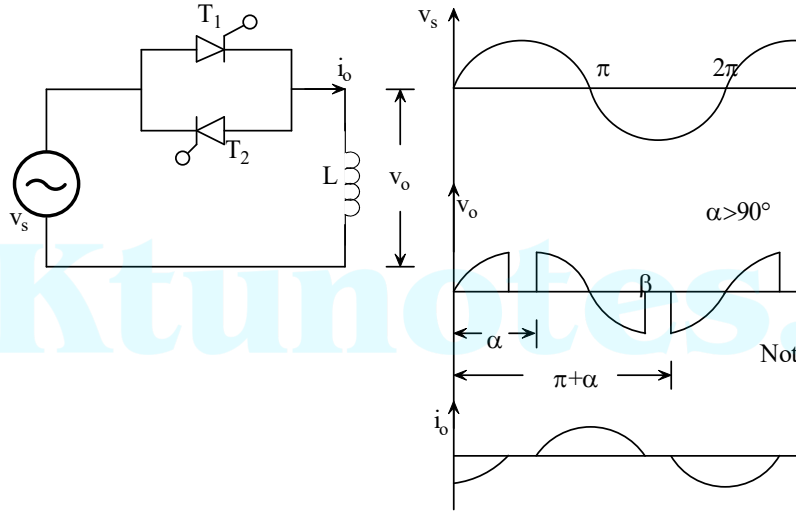
$$\text{RMS value of output voltage, } V_o = \sqrt{\frac{1}{\pi} \int_{\alpha}^{\pi} V_m^2 \sin^2 \omega t d\omega t} = V_m \sqrt{\frac{1}{2\pi} (\pi - \alpha + \frac{\sin 2\alpha}{2})}$$

When $\alpha = 0$, $V_o = \frac{V_m}{\sqrt{2}} = V_s$ and when $\alpha = \pi$, $V_o = 0$.

By varying $\alpha = 0$ to π , RMS value of output voltage V_o can be controlled from V_s to 0.

$$\text{RMS value of load current, } I_o = \frac{V_o}{R}$$

L LOAD



During positive half cycle, T_1 is triggered at $\alpha \geq \pi/2$. When T_1 is conducting, v_o is v_s . At the end of the positive half cycle, T_1 will continue to conduct upto β ($\beta = \text{extinction angle} = 2\pi - \alpha$) due to inductance. (Note : Average voltage across inductance is zero; equal positive & negative voltage). T_2 is triggered at $\pi + \alpha$. When T_2 is conducting, $v_o = -v_s$. T_2 conducts upto β . For a voltage controller with L load,

$$L \frac{di}{dt} = V_m \sin \omega t \text{ where } \alpha \leq \omega t \leq \beta$$

$$\begin{aligned} i &= \frac{1}{L} \int V_m \sin \omega t d\omega t \\ &= -\frac{V_m}{\omega L} \cos \omega t + K \end{aligned}$$

$$\text{At } \omega t = \alpha, i = 0$$

$$\text{Hence, } -\frac{V_m}{\omega L} \cos \alpha + K = 0$$

$$K = \frac{V_m}{\omega L} \cos \alpha$$

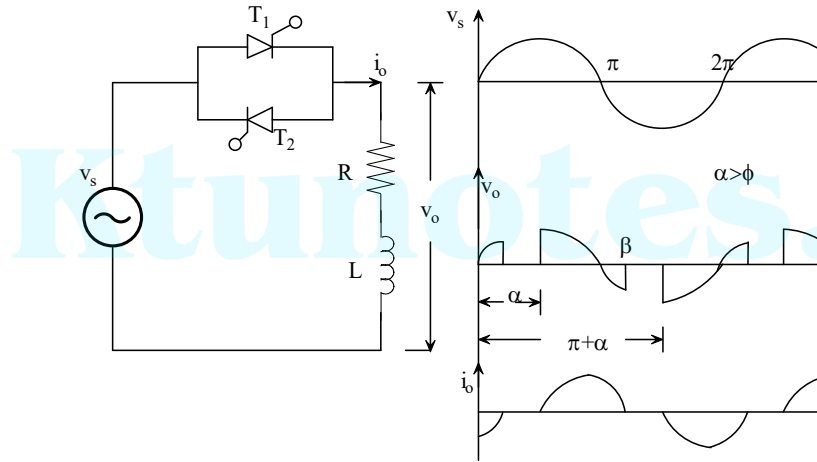
$$i_o(t) = \frac{V_m}{\omega L} (\cos \alpha - \cos \omega t)$$

$$\text{If } \alpha = \pi / 2,$$

If $\alpha = \pi / 2$, output voltage and input voltage waveforms are same. Load is directly connected to ac source.

Hence, for L load, output voltage can be controlled only if $\alpha \geq \pi / 2$.

RL LOAD



During positive half cycle, at $\omega t = \alpha$, T_1 is triggered. When T_1 is conducting, $v_o = v_s$. During negative half cycle, T_2 will continue to conduct upto β (where $\beta =$ extinction angle; $\beta > \pi$). The current i_o continues to flow due to the stored energy in the inductance. T_2 is triggered at $\pi + \alpha$. When T_2 is conducting, $v_o = -v_s$ upto $\pi + \beta$.

If $\alpha \leq \Phi$, circuit behaves as if RL load is directly connected to ac source.

SEQUENCE CONTROL OF AC VOLTAGE CONTROLLERS

For improving the input power factor and for reduction of harmonics in the line current, sequence control of ac regulators is employed. Two or more voltage controllers are connected in parallel and they are triggered in proper sequence to obtain output with low harmonics.

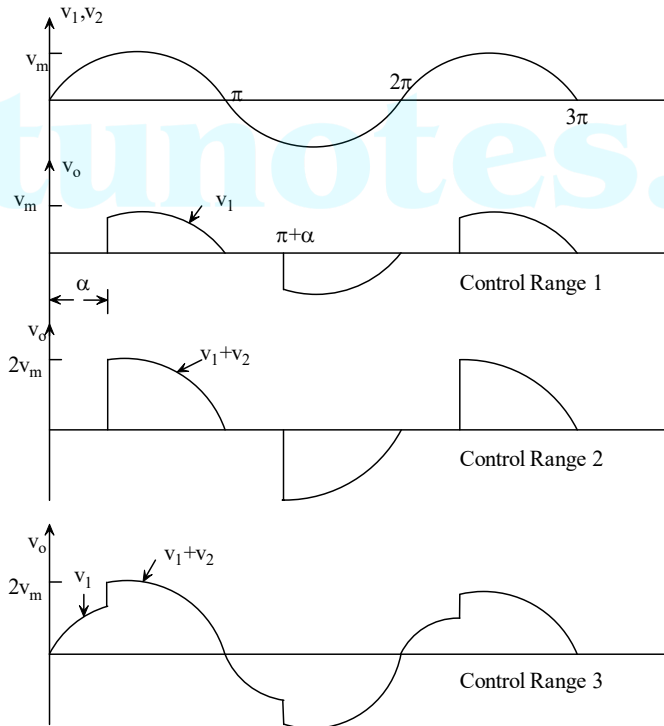
TWO STAGE SEQUENCE CONTROL

Let $v_1 = v_2 = V_m \sin \omega t$ and $v_1 + v_2 = 2V_m \sin \omega t$. Let V_1 and V_2 be the RMS values of v_1 and v_2 . Assume that the load is pure resistive.

Control Range 2 : $0 \leq V_o \leq (V_1 + V_2)$

T_3 & T_4 are turned OFF and T_1 & T_2 are operated as voltage controllers.

RMS value of output voltage, $V_o = (V_1 + V_2) \sqrt{\frac{1}{\pi} (\pi - \alpha + \frac{\sin 2\alpha}{2})}$; $0 \leq \alpha \leq \pi$



Control Range 3 : $V_1 \leq V_o \leq (V_1 + V_2)$

At $\omega t = 0$ to α , T_3 is turned ON; $v_o = v_1$.

At $\omega t = \alpha$, T_1 is turned ON; T_3 is reverse biased due to v_2 and hence it is turned OFF.

At $\omega t = \pi$, T_1 is line commutated; T_4 is turned ON by applying a gate pulse; $v_o = v_2$.

At $\omega t = \pi + \alpha$, T_2 is turned ON; T_4 is reverse biased due to v_2 and hence it is turned OFF.

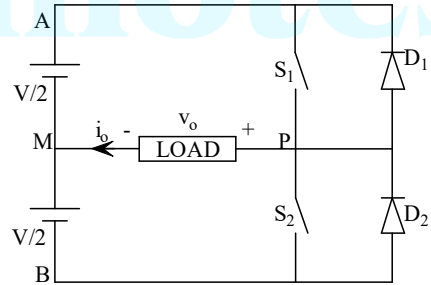
At $\omega t = 2\pi$, T_2 is line commutated.

Also, voltage source inverters may further be classified as a) square-wave inverter and b) pulse-width modulated (PWM) inverter.

- Square-Wave Inverter – produces a square wave ac voltage of constant magnitude. The output voltage of this type of inverter can only be varied by controlling the input voltage.
- Pulse-Width Modulated (PWM) Inverters – the output has one or more pulses per cycle. By varying the width of these pulses, the output voltage can be controlled. In PWM-VSI, the dc input voltage is essentially constant.

VOLTAGE SOURCE INVERTER

1-PHASE HALF-BRIDGE INVERTER



S_1 and S_2 are two controlled power semiconductor devices (power transistor, power MOSFET, etc) which act as switches.

R-LOAD

At $\omega t=0$, S_1 is turned ON by applying a base or gate current signal. When S_1 is ON, terminal P is connected to positive terminal A of dc source. Load voltage is V . When the gate signal is removed from S_1 and it is turned OFF. At the same time, the gate signal is applied to S_2 and it is turned ON. When S_2 is conducting, load terminal P is connected to negative terminal B of dc source. Load voltage is $-V/2$.

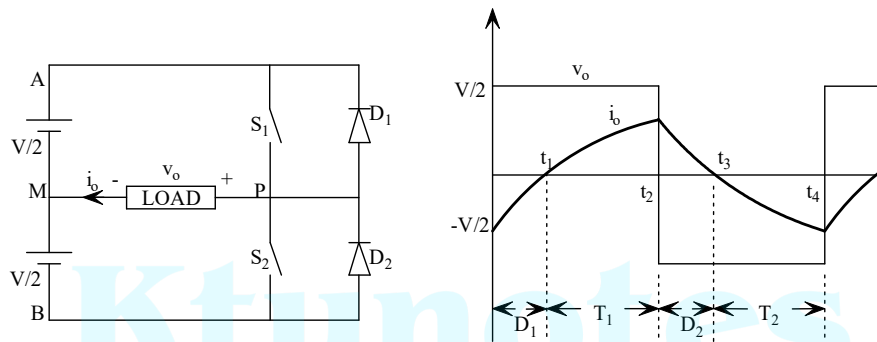
Hence, by closing S_1 and S_2 alternately for half of time periods, a square-wave voltage is produced at the output.

Since conduction period of any switch is equal to duration of its base/gate current, the output load voltage is controlled by control signal of the inverter.

For a pure resistive load, i_o and v_o will be in phase and of same wave shape.

For pure resistive load, the diodes D_1 and D_2 do not play any role.

voltage is reversed. Output voltage is similar to that with R-load.



Mode 1 ($t_1 < t < t_2$)

At t_1 , S_1 is turned ON.

$v_o = V/2$; load current i_o increases from zero.

At $t=t_2$, S_1 is turned OFF. Current reaches peak; energy is stored in the inductance since both v_o and i_o are positive.

Mode 2 ($t_2 < t < t_3$)

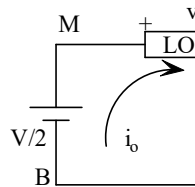
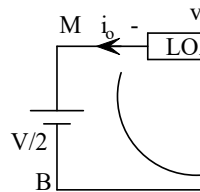
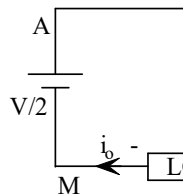
At t_2 , even though S_1 is turned OFF, load current will continue to flow in the same direction through D_2 due to stored energy in the inductance. $v_o = -V/2$; i_o decreases; stored energy is fed back to lower half of the source.

Mode 3 ($t_3 < t < t_4$)

At t_3 , $i_o=0$; S_2 is turned ON, $v_o = -V/2$; load current is negative.

At t_4 , load current reaches peak.

Mode 4 ($t_4 < t < t_5$) or ($0 < t < t_1$)

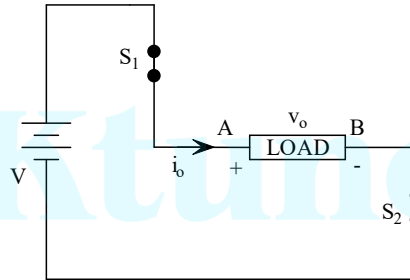


RL load

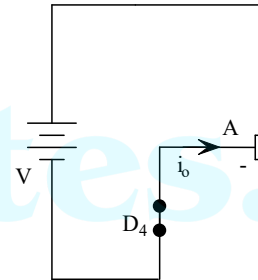
Mode 1

At $t=t_1$; S_1 & S_2 are turned ON. $v_o = +V$; i_o starts increasing from zero. i_o reaches

MODE 1 ($t_1 < t < t_2$)



MODE 2 ($t_2 < t < t_3$)



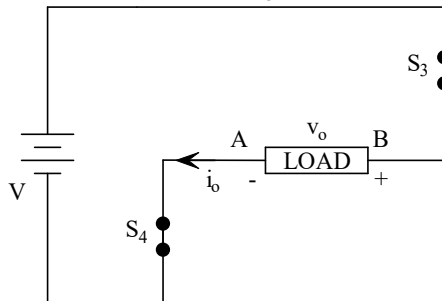
Mode 2

At $t=t_2$; S_1 & S_2 are turned OFF. i_o starts decreasing. Stored energy in the inductor flows through D_3 and D_4 . $v_o = -V$.

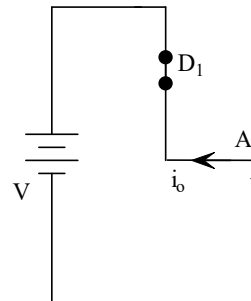
Mode 3

At $t=t_3$; load current becomes zero; S_3 & S_4 are turned ON; i_o flows in negative direction.

MODE 3 ($t_3 < t < t_4$)



MODE 4 ($t_4 < t < t_5$)

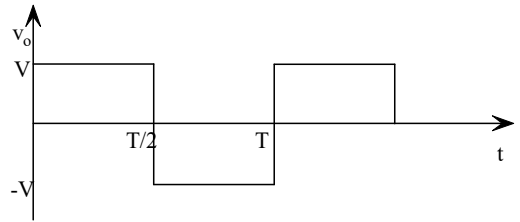


Mode 4

At $t=t_4$; load current becomes maximum in reverse direction; S_3 & S_4 are turned OFF. Stored energy in the inductor flows through D_1 & D_2 due to stored energy. $v_o = +V$.

Special Case : Full bridge inverter with **L** load

ANALYSIS OF 1-PHASE FULL-BRIDGE INVERTER



The square wave has half-wave symmetry and odd symmetry.

$$b_n = \frac{4}{2\pi} \int_0^{\pi} V \sin n\omega t d\omega t = \frac{4V}{2\pi n} [-\cos n\omega t] = \frac{4V}{n\pi} \text{ where } n = 1, 3, 5, \dots$$

$$v_o(t) = \sum_{n=1,3,5,\dots}^{\infty} \frac{4V}{n\pi} \sin n\omega t$$

$$v_1(t) = \frac{4V}{\pi} \sin \omega t$$

$$\text{RMS value of fundamental output voltage, } V_1 = \frac{2\sqrt{2}V}{\pi} = 0.9V$$

$$\text{RMS value of output voltage, } V_o = \sqrt{\frac{1}{\pi} \int_0^{\pi} V^2 d\omega t} = V$$

$$\text{Total Harmonic Distortion, } THD = \frac{\sqrt{V_o^2 - V_1^2}}{V_1} = 0.4843$$

Note : RMS value of fundamental, 3rd, 5th, 7th and 9th harmonics output voltage
 $V_1=0.9V$, $V_3=0.3V$, $V_5=0.18V$, $V_7=0.13V$, $V_9=0.1V$.

3-PHASE BRIDGE INVERTER (SIX STEP INVERTER)

A 3-phase inverter is formed by connecting three 1-phase half-bridge inverters.

A 3-phase inverter is classified as : a) 180° conduction mode inverter and b) 120° conduction mode inverter. This classification is based on the period of conduction of each switch.

positive terminal of the source and load terminal B is connected to negative terminal of the source. Equivalent circuit is shown below. Resistance between positive terminal A and neutral N is R . Resistance between negative terminal B and neutral N is R .

Hence, load voltages are

$$V_{AN} = \frac{V}{3}, V_{BN} = -\frac{2V}{3}, V_{CN} = \frac{V}{3}$$

Line voltages are

$$V_{AB} = V_{AN} - V_{BN} = V$$

$$V_{BC} = V_{BN} - V_{CN} = -V$$

$$V_{CA} = V_{CN} - V_{AN} = 0$$

Similarly, other equivalent circuits and voltages can be known.

0-60° ↔ T₁, T₅, T₆

$$V_{AN} = \frac{V}{3}$$

$$V_{BN} = -\frac{2V}{3}$$

$$V_{CN} = \frac{V}{3}$$

$$V_{AB} = V_{AN} - V_{BN} = V$$

$$V_{BC} = V_{BN} - V_{CN} = -V$$

$$V_{CA} = V_{CN} - V_{AN} = 0$$

120-180° ↔ T₁, T₂, T₃

$$V_{AN} = \frac{V}{3}$$

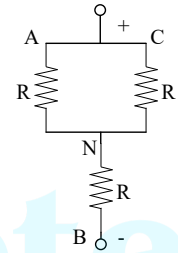
$$V_{BN} = \frac{V}{3}$$

$$V_{CN} = -\frac{2V}{3}$$

$$V_{AB} = V_{AN} - V_{BN} = 0$$

$$V_{BC} = V_{BN} - V_{CN} = V$$

$$V_{CA} = V_{CN} - V_{AN} = -V$$



60-120° ↔ T₁, T₂, T₆

$$V_{AN} = \frac{2V}{3}$$

$$V_{BN} = \frac{-V}{3}$$

$$V_{CN} = -\frac{V}{3}$$

$$V_{AB} = V_{AN} - V_{BN} = V$$

$$V_{BC} = V_{BN} - V_{CN} = 0$$

$$V_{CA} = V_{CN} - V_{AN} = -V$$

180-240° ↔ T₂, T₃, T₄

$$V_{AN} = -\frac{V}{3}$$

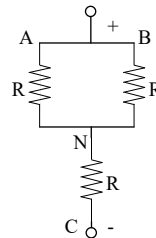
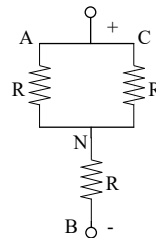
$$V_{BN} = \frac{2V}{3}$$

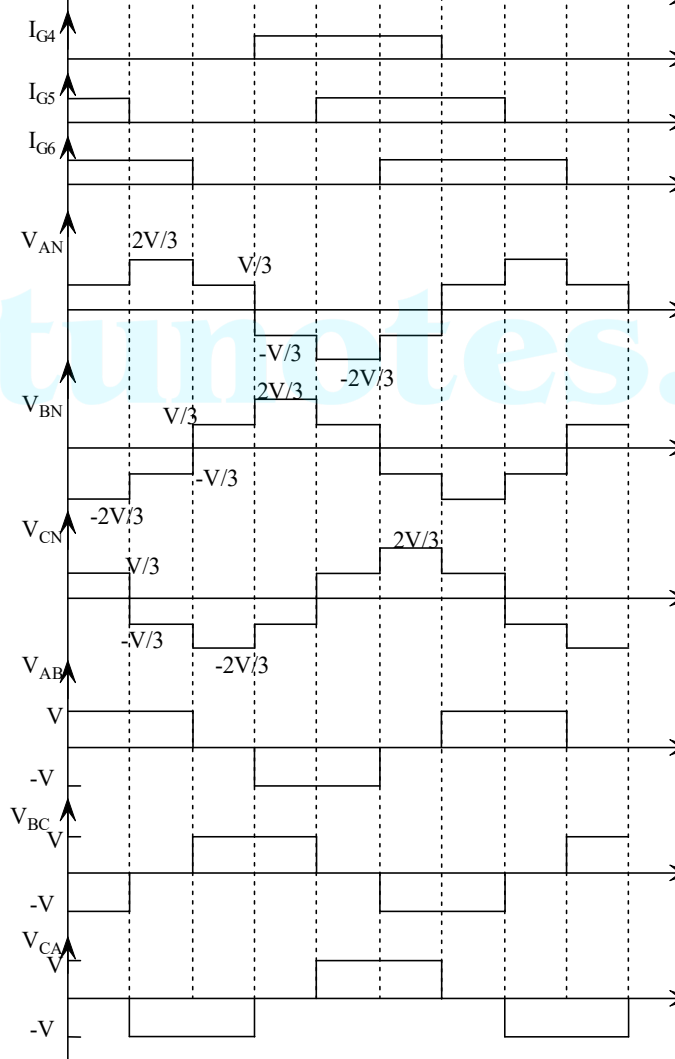
$$V_{CN} = -\frac{V}{3}$$

$$V_{AB} = V_{AN} - V_{BN} = -V$$

$$V_{BC} = V_{BN} - V_{CN} = V$$

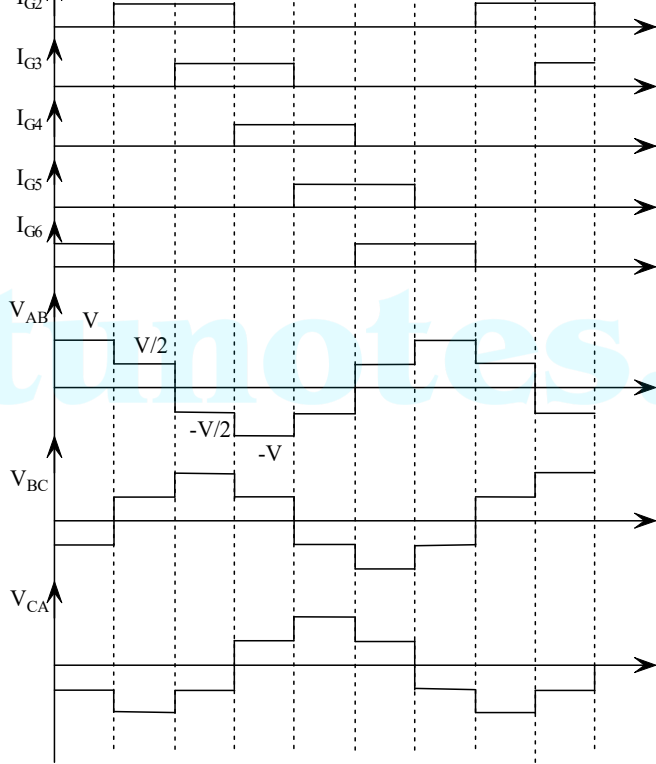
$$V_{CA} = V_{CN} - V_{AN} = 0$$





120° CONDUCTION

In 120° conduction scheme, each device conducts for 120°. It is preferable for because it provides a six step waveform across any phase. As each device cond devices are in conduction state at any instant.

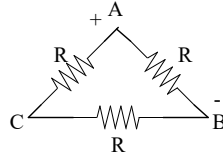


0-60° ↔ T₁, T₆

$$V_{AB} = V$$

$$V_{BC} = -\frac{V}{2}$$

$$V_{CA} = -\frac{V}{2}$$



60-120° ↔ T₁, T₂

$$V_{AB} = \frac{V}{2}$$

$$V_{BC} = \frac{V}{2}$$

$$V_{CA} = -V$$

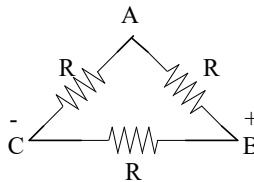
C-

120-180° ↔ T₂, T₃

$$V_{AB} = -\frac{V}{2}$$

$$V_{BC} = V$$

$$V_{CA} = -\frac{V}{2}$$



180-240° ↔ T₃, T₄

$$V_{AB} = -V$$

$$V_{BC} = \frac{V}{2}$$

$$V_{CA} = \frac{V}{2}$$

C+

devices of the same leg.

Utilization factor of 180° conduction scheme is 33% whereas for 120° conduction scheme it is 28.9%.

If thyristors are used, in 180° conduction scheme, complementary voltage conduction angle is required to turn OFF the two thyristors of same leg. In 120° conduction mode, anti-parallel diodes are required.

VOLTAGE CONTROL IN 1-PHASE INVERTER PULSE WIDTH MODULATION

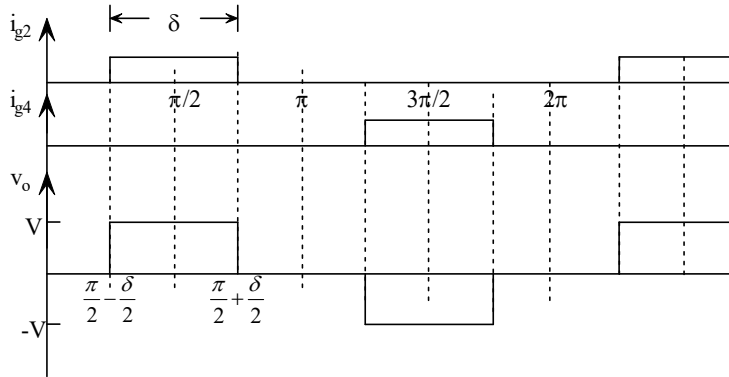
PWM (Pulse Width Modulation) is the most suitable method of voltage control in a 1-phase inverter. It controls the voltage but also reduces the harmonic content.

Elimination of lower order harmonics results in reduced size of filter which saves space. Different PWM techniques are i) single pulse width modulation ii) Multiple-pulse width modulation and iii) sinusoidal PWM.

SINGLE PULSE WIDTH MODULATION

There is only one pulse per half-cycle and the width of the pulse is varied to control the output voltage.

$$\text{RMS output voltage, } V_o = \sqrt{\frac{2}{2\pi} \int_{\frac{\pi-\delta}{2}}^{\frac{\pi+\delta}{2}} V_s^2 d\omega t} = V_s \sqrt{\frac{\delta}{\pi}}$$



By Fourier analysis,

Output voltage wave has quarter-wave symmetry. [Half-wave symmetry (n is odd) and a_0 and $a_n=0$].

By Fourier analysis,

$$\text{Modulation Index} = \frac{\text{Amplitude of reference signal}}{\text{Amplitude of carrier signal}} = \frac{A_r}{A_c}$$

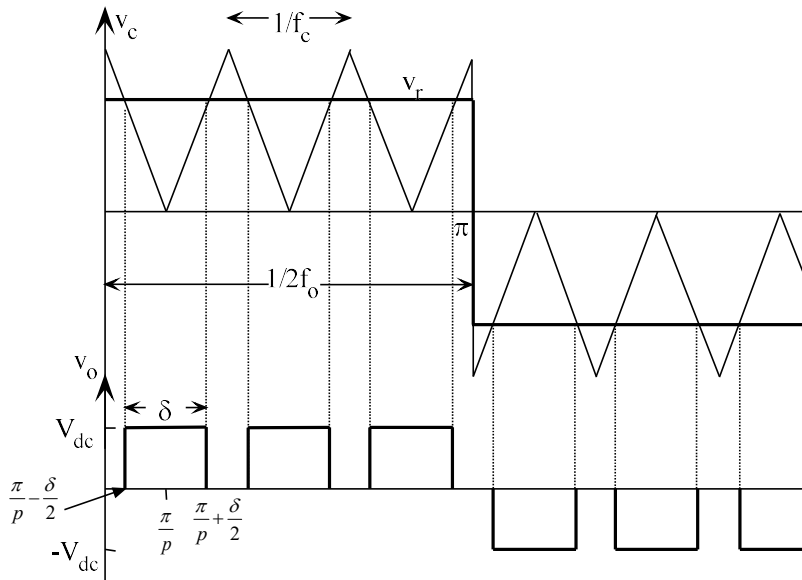
Modulation index m varies from 0 to 1.

Frequency Modulation ratio $m_f = \frac{f_c}{f_o}$ (m_f indicates the number of pulses per cycle)

No. of pulses per second = f_c

No. of pulses per half cycle = $p = f_c \times \frac{T_o}{2} = \frac{f_c}{2f_o}$

$$\text{RMS output voltage, } V_o = \sqrt{\frac{p}{\pi} \int_{\frac{\pi}{p}}^{\frac{\pi}{p} + \frac{\delta}{2}} V_{dc}^2 d\omega t} = V_{dc} \sqrt{\frac{p\delta}{\pi}}$$

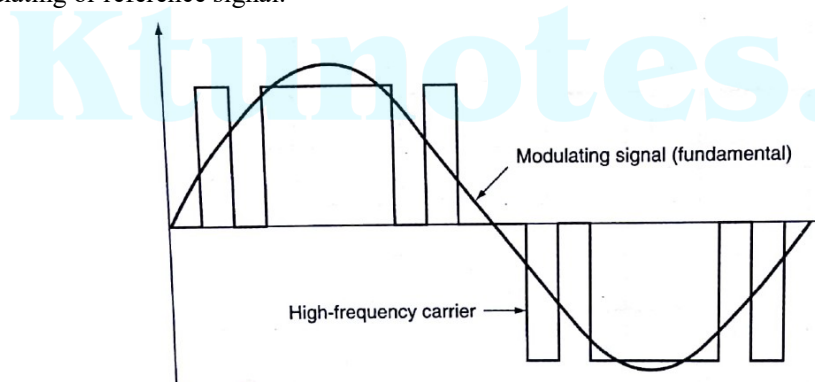


some higher order harmonics would increase. But, such higher order harmonics can be filtered out by using a low pass filter.

Since the modulating or reference wave has a square shape, this PWM technique is called square wave PWM.

SINUSOIDAL PULSE WIDTH MODULATION

In the sinusoidal pulse-width modulation technique, the pulse width of a high-frequency carrier signal, called the carrier, is varied in accordance to the amplitude of a low-frequency modulating or reference signal.



Gating signals are generated by comparing a sinusoidal reference signal v_{sine} (of frequency f_o) with a triangular carrier wave v_{tri} of frequency f_c .

$$\text{Amplitude modulation ratio, } m_a = \frac{\text{Amplitude of reference voltage}}{\text{Amplitude of carrier signal}} = \frac{A_{\text{sine}}}{A_{\text{tri}}}$$

Amplitude of carrier wave is generally kept constant.

$$\text{Frequency modulation ratio, } m_f = \frac{\text{Frequency of triangular wave}}{\text{Frequency of sine wave}} = \frac{f_c}{f_o}$$

PWM WITH BIPOLAR SWITCHING

When $v_{\text{sine}} > v_{\text{tri}}$, S_A is ON (S_A' is OFF) and S_B is OFF (S_B' is ON), $v_o = +V_{dc}$

When $v_{\text{sine}} < v_{\text{tri}}$, S_A is OFF (S_A' is ON) and S_B is ON (S_B' is OFF), $v_o = -V_{dc}$

Note: Switching signals to two switches in a leg will be complimentary (When one is ON, the other is OFF) and these two switches are never off simultaneously.

The output voltage switches between $-V_{dc}$ and $+V_{dc}$ voltage levels. Hence, bipolar voltage switching.

Peak of the fundamental output voltage, $\hat{V}_{o1} = m_a V_{dc}$ where $m_a \leq 1$

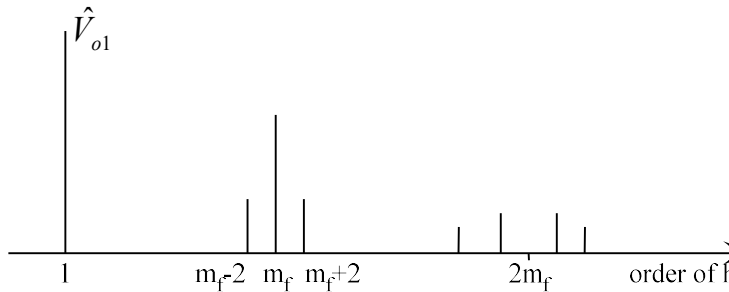
Peak (RMS) value of the fundamental component of the output voltage modulation index m_a .

(Note : Peak value of the fundamental output voltage with square wave output,

When $m_a > 1$, square wave operation,

$$V_{dc} < \hat{V}_{o1} < \frac{4V_{dc}}{\pi}$$

The harmonics in the inverter output voltage waveform appear as side the switching frequency and its multiples, that is, around harmonics m_f , $2m_f$ general pattern holds true for all values of m_a in the range 0-1.



The harmonic m_f should be an odd integer. Choosing m_f as an odd in symmetry as well as half-wave symmetry. Therefore, only odd harmonics are harmonics are absent. Moreover, only sine terms are present.

Generalised harmonics of output voltage v_o for a large m_f

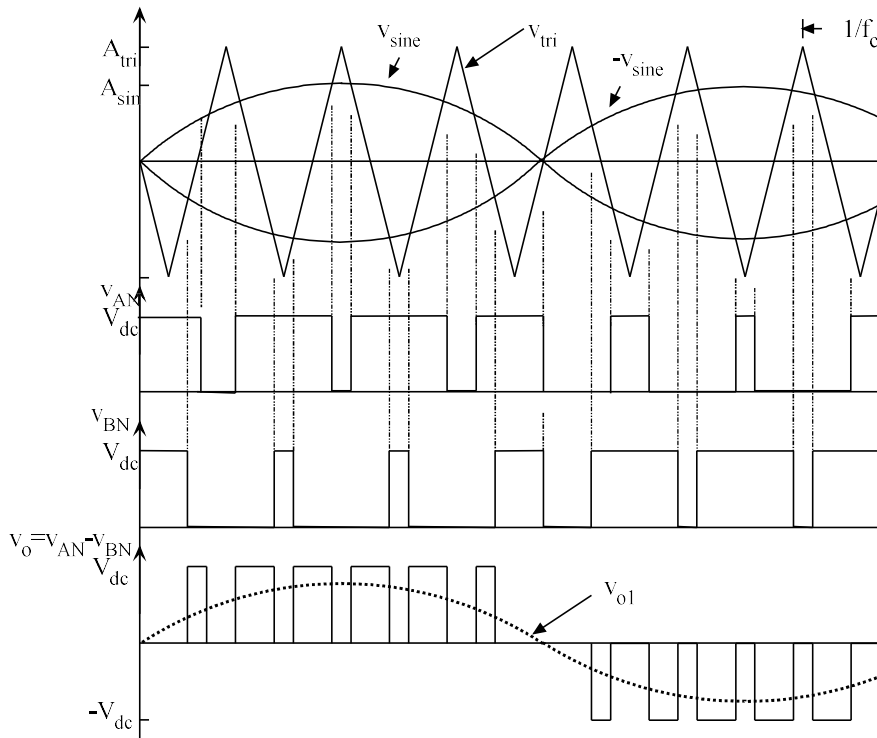
Order of harmonic n	$\frac{\hat{V}_{on}}{V_{dc}}$ with $m_a = 0.8$	$\frac{\hat{V}_{on}}{V_{dc}}$ with $m_a = 1$
1	0.8	1
m_f	0.82	0.6
$m_f \pm 2$	0.22	0.32
$2m_f \pm 1$	0.31	0.18
$2m_f \pm 3$	0.14	0.21

It is desirable to use as high a switching frequency as possible for eliminating harmonics. Higher order harmonics can be easily filtered out using a low pass filter. As the switching frequency is increased, switching losses may increase and hence inverter efficiency decreases.

If $m_a > 1$ leads to over-modulation which causes the output voltage to contain many more harmonics. Overmodulation causes the output voltage to contain many side-bands as compared with the linear range (with $m_a < 1$). Hence, over-modulation is avoided in applications requiring low distortion.

PWM WITH UNIPOLAR SWITCHING

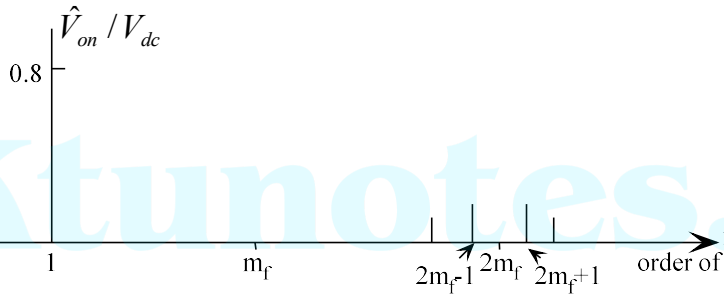
In PWM with unipolar voltage switching, the switches in the two legs of the full-bridge are switched simultaneously. Here, the legs A and B of the full-bridge inverter are driven by comparing v_{tri} with v_{sine} and $-v_{sine}$ respectively.



Also, the current ripples are less with unipolar switching.

Here, in unipolar switching scheme m_f is chosen to be even whereas in bipolar it is odd.

Note also only odd harmonics are present.



CURRENT SOURCE INVERTER

LOAD COMMUTATED CURRENT SOURCE INVERTER

In the voltage source inverter (VSI), the load voltage is independent of the type of source inverter (CSI), the load current is independent of the load, however, it depends upon the type of the load. A voltage source is converted into a current source by a large series inductor. Because of the series inductor, variation in the source current ripple is negligible.

Load commutation is possible only for a leading power factor load. A load is commutated by the negative voltage across the load when other pair of thyristors is forward biased. In the case of R or RL loads, a capacitor is connected in parallel across the load.

